

Large Language Models as Decision-Makers Under Uncertainty: Value of Information, Error Decomposition and an Extract-then-Solve Guardrail

Chiheb Mejrissi*

Department of Computer Science, Example University, 1 Example Street, Example City, Country

Abstract

[Intent: Problem (decisions under uncertainty handed to LLMs); the benchmark with exact Bayes-optimal actions on open-weight models; the two headline questions, value-of-information pricing (C5) and where the loss arises (C6); the guardrail (C4, C4b); the replications (C1, C2(b)); outcomes once recorded.]

Keywords: large language models, decision support, value of information, Bayesian decision theory, calibration, guardrails, benchmark

1. Introduction

[Intent: Motivate LLMs as decision aids in costly-test, triage, allocation and updating problems; state the gap (G1–G4 of the literature map); lead with C5 (do models price information?) and C6 (is the loss misreading, inference or action?); present C4 and C4b as the design answer; state C1 and C2(b) as replications of prior work on multi-action families and pinned open weights; list the contributions and the roadmap.]

2. Related work

[Intent: Position against LLM rationality and decision benchmarks (closest: Sun et al. 2026, exact regret with a belief/optimization split on binary tasks;

*Corresponding author.

Email address: chiheb.mejrissi@example.org (Chiheb Mejrissi)

Yamin et al. 2026, belief–action coherence), heuristics and biases in LLMs, calibration of stated probabilities, framing and order sensitivity, program-of-thought and tool use, and value of information in decision analysis; say what each covers and what Paper 8 adds.]

3. Benchmark: decision problems with exact optimal actions

The benchmark consists of decision problems stated in text whose Bayes-optimal action is computed exactly by a solver. Exactness is the design principle: it lets every answer be scored by its expected loss against the optimum, with no human or model judge, and it lets the same item be restated in ways that provably leave every expected loss unchanged. The generators, solvers, strata, variants and checks described in this section were fixed before any model was run.

3.1. Task families and exact solvers

Each item presents a decision-maker with two to several named actions, the probabilities and costs needed to evaluate them, and a short story that gives the problem a setting. Four families cover the decisions the paper is about: whether to pay for information before acting, where to set an action threshold when errors cost different amounts, how much to commit against uncertain demand, and how to update a belief from evidence. Each family has an exact solver and a named *trap heuristic*, a plausible shortcut that some items are built to make wrong (Section 3.2).

F1: verification and stopping. Two causes H_1 and H_0 are possible, with prior $p = \Pr(H_1)$. The decision-maker may act for H_1 , act for H_0 , or first buy a test of cost c with sensitivity s and specificity t and then act on its result. Acting for H_0 when H_1 holds costs a miss M ; acting for H_1 when H_0 holds costs a false alarm A . In sub-family F1a ([TODO: param: F1a items] items) a single test is available and the solver compares the expected loss of acting now with that of testing and then acting optimally on the result. In F1b ([TODO: param: F1b items] items) up to $T \in$ [TODO: param: F1b test counts] conditionally independent tests can be bought in sequence, each at cost c , and the solver computes the optimal policy by backward induction. The optimal first action is to test exactly when the value of the test’s information exceeds its cost [1]. F1 is therefore the family on which information pricing is measured (Section 5.1). Trap heuristic: act for the more probable cause, ignoring the costs and the test.

F2: asymmetric-cost triage. A case has prior p of being positive and one test result with sensitivity s and specificity t . A missed positive costs C_{FN} and a needless intervention costs C_{FP} . The Bayes action is to intervene if and only if the posterior exceeds $C_{FP}/(C_{FP} + C_{FN})$, the classical treatment threshold [2]. Trap heuristic: intervene if the posterior exceeds one half, ignoring the cost asymmetry.

F3: allocation under uncertain demand. In F3a ([TODO: param: F3a items] items), a newsvendor faces a discrete demand on [TODO: param: F3a support points] support points with stated probabilities, a unit price, a unit cost and a salvage value, and chooses an order quantity among the support points. The solver returns the smallest quantity whose cumulative probability reaches the critical fractile. In F3b ([TODO: param: F3b items] items), a fixed stock is split between two sites with independent discrete demands, and the solver enumerates the splits. Trap heuristic: order, or split by, the mean demand rounded to the nearest support point. F3 asks for no probability, only an action.

F4: Bayesian updating. A hypothesis H_1 has prior p , and [TODO: param: F4 evidence count] conditionally independent test results or urn draws with stated likelihoods are observed. The decision-maker states the posterior and then accepts or declines a bet that pays X if H_1 holds and loses Y otherwise; the Bayes action accepts if and only if the exact posterior p^* satisfies $p^*X > (1 - p^*)Y$. Trap heuristic: base-rate neglect, the posterior computed from the likelihood ratio alone.

3.2. Regret, strata and decision-equivalent variants

Normalized regret. Let $L(a)$ be the exact expected loss of action a on an item, $L^* = \min_a L(a)$ and $L_{\max} = \max_a L(a)$ over the item's actions. The normalized regret of a is

$$r(a) = \frac{L(a) - L^*}{L_{\max} - L^*} \in [0, 1], \quad (1)$$

which is zero for the Bayes action and one for the worst listed action, so items with different stakes and currencies are comparable. A model is *practically optimal* in a family if its mean normalized regret is below $\delta =$ [TODO: param: delta], a margin fixed before any run.

Separation. An item is kept only if the second-best action’s expected loss exceeds the best by at least [TODO: param: separation margin] ($L_{\max} - L^*$). Near-ties, on which a small numerical slip would change the optimal action at almost no cost, are thereby excluded, and every wrong action carries a regret bounded away from zero.

Trap and non-trap strata. An item is a *trap* item if its family’s heuristic gives a wrong action on it and a *non-trap* item if the heuristic gives the Bayes action. Each family is drawn in equal halves from the two strata by rejection sampling on the heuristic. The strata are defined by construction (check K1, Section 4.3): each heuristic has positive regret on every trap item and zero regret on every non-trap item. A model that follows a heuristic therefore loses exactly on the trap stratum, and the difference between strata locates the loss.

Balance of the optimal action. Two families are further balanced on their Bayes action, because a family in which one action is nearly always optimal rewards a model that always gives it. In F4, the prior is drawn on both sides of one half, so that base-rate neglect overstates the posterior in some items and understates it in others, and each consecutive block of four items contains one item of each (stratum, Bayes action) cell. In F1, each block of four contains one item of each (stratum, test is the optimal first action) cell, so that half of the F1 items call for buying the test and half do not, within each stratum. This balance is what gives the value-of-information measure of Section 5.1 equal cells. If a cell cannot be filled within the generator’s draw limit, F1 is balanced on the test action alone and the trap share per test cell is reported. F2 and F3 are not rebalanced; the regret of every constant-action policy is reported beside each model’s instead (Section 5.4).

Decision-equivalent variants and the noise floor. A *decision-equivalent variant* of an item is a restatement that leaves its vector of expected losses unchanged, so that any change of action between the canonical item and the variant is an error. Three transformations are used: gain/loss framing (in F1–F3 the same outcomes are stated as amounts lost or as amounts kept; in F4 the bet is stated on H_1 or on its complement), reversal of the option order, and paraphrase, with [TODO: param: paraphrase templates] templates written before any run (the first being canonical). Equivalence is checked numerically on every item (check K3, tolerance [TODO: param: K3 tolerance]). The *flip rate* of a transformation is the share of items whose action

differs between the canonical item and its variant. It is compared with the model’s *noise floor*, the rate at which the action changes when the identical prompt is submitted again (Section 4.2).

3.3. Item generation and contamination control

Item parameters are drawn from laws fixed in advance. The prior p is log-uniform on [TODO: param: prior range] (in F4, reflected about one half with equal probability); sensitivities and specificities are uniform on [TODO: param: sensitivity/specificity range] and rounded to two decimals; cost ratios (M/A , C_{FN}/C_{FP} , X/Y , price over cost) are log-uniform on [TODO: param: cost-ratio range] and costs are integers in [TODO: param: cost range]. Each item is set in one of [TODO: param: story domains] surface domains (clinical testing, IT incidents, quality inspection and logistics), drawn uniformly. Probabilities are shown as percentages, with fixed units and currency.

Because the items are generated rather than collected, no item can have appeared in a model’s training data, and the surface stories and numbers are new. Each family has [TODO: param: items per family] items, drawn from a seed block ([TODO: param: seed block]) registered before generation and disjoint from every other seed in the project. The SHA-256 hash of each item file is committed to the public repository before the first model call, and the item files themselves are released only after all runs end, so the items cannot leak into a model or into prompt design while the study runs. The conditions run on a subset use the first [TODO: param: shared prefix per family] items of each family in its stratified order, which preserve the strata and the balance cells. Separately, a *canonical-story control* of [TODO: param: canonical-story items per family] famous textbook wordings per family is run under direct prompting only and reported apart, to show the size of any contamination effect rather than hide it.

4. Models and decision conditions

4.1. Models

The large language models (LLMs) in this study are the object of research. They are open-weight instruction models, each downloaded once at a pinned repository revision, verified by the SHA-256 hash of its weight file and served locally by an open-source inference server; no commercial API is used. Pinned weights make the tested models immutable: the results below refer to exactly these files, and anyone can rerun them. The paper states

Table 1: Model panel: developer, repository, revision, weight-file hash, quantization, licence and inference server, as pinned before the first model call.

[TODO: param: panel table built from the lab’s pins.json]

each model’s repository, revision, quantization and server version (Table 1) and claims nothing about closed-API models, other models or later versions.

The panel has [TODO: param: panel size] models, each from a different developer, chosen before any run to vary developer, architecture and size, and restricted to licences that permit research use and weights that can be downloaded without an access request: a Llama instruction model (Meta, from an ungated conversion of the released weights), a Qwen3 mixture-of-experts instruction model (Alibaba), the reasoning-native gpt-oss model (OpenAI), an OLMo instruction model (Ai2), whose training data are public, a Granite instruction model (IBM) and Phi-4-mini-instruct (Microsoft). Substitutes for each candidate were fixed in advance, for use only if a candidate could not be pinned or loaded; any substitution is recorded before that model’s first call. No model is added after the first result. With [TODO: param: panel size] models no trend with model size can be separated from developer and architecture, and none is claimed.

Decoding. Every model answers every condition with greedy decoding (temperature zero, no nucleus truncation and no repetition, frequency or presence penalty), from its own chat template, with a fixed maximum number of output tokens per condition (Appendix Appendix A). The reasoning-native model runs in its native response format at its lowest reasoning setting in every condition, including the direct one; its reasoning text is logged and never parsed. Server settings (version, quantization, context length, threads and batching) are pinned and recorded. Under greedy decoding the only remaining variation between identical requests is server nondeterminism, and the noise floor measures exactly that: [TODO: param: noise-floor items per family] items per family are resubmitted in a fresh server process with a different request order. This is the variation a user of the same deployment would see.

4.2. Conditions

All conditions share one system prompt that casts the model as an adviser to a decision-maker and asks for a single JSON object. Where the family

has a probability step (F1, F2, F4), the object holds the probability of H_1 after the evidence and then the action; in F3 it holds the action only. The conditions differ in what the model is asked to produce and in who computes the decision (Table 2). Prompts, schemas and formula sentences were written before any run and were not tuned on model outputs; they are given in Appendix A.

Direct (P0). The model reads the item and answers with the JSON object only. P0 is the reference condition, and it is the condition on which the decision-equivalent variants and the canonical-story control are run.

Free reasoning (P1). The model is asked to think step by step before giving the JSON object on its last line [TODO: cite: Kojima et al. 2022 (L14), NeurIPS; bib entry pending a venues.csv row]. P1 is the strongest condition in which the model still does everything itself.

Structured direct (P2). In a single answer, the model returns its reading of the item’s quantities $\hat{q}$ in the family’s extraction schema (the same schema as the guardrail’s), then its stated probability $\hat{p}$, then its action $\hat{a}$. Because the reading, the probability and the action are all observed, P2 makes it possible to locate where a decision’s loss arises (Section 5.2). P2’s own regret is reported beside P0’s.

Program-of-thought (PoT). The model writes a Python function `decide()` that returns an action label, and the harness executes it [TODO: cite: program-aided and program-of-thought prompting (L26); bib entries pending]. Execution is sandboxed: a separate process with no network or file-system access, a CPU limit of [TODO: param: PoT CPU limit], and only the standard-library modules `math`, `fractions` and `itertools` importable. PoT is run on the canonical item and on its framed variant. It is the strongest baseline in which computation is delegated to code while the model still writes the model of the problem.

Formula-given (FG). The direct condition with one added sentence per family stating the optimal decision rule in words: compare the expected loss of acting now with that of testing (F1); intervene if the posterior exceeds $C_{FP}/(C_{FP} + C_{FN})$ (F2); order the smallest quantity whose cumulative probability reaches the critical fractile, or enumerate the splits (F3); accept if the posterior times X exceeds its complement times Y (F4). FG separates

Table 2: Decision conditions. “Reads” is what the model extracts, “states” what it outputs, “decides” who selects the action.

Condition	Model outputs	Action chosen by	Also run on
P0 direct	probability ^a , action	model	framing, order, p
P1 free reasoning	reasoning, probability ^a , action	model	—
P2 structured direct	quantities, probability ^a , action	model	—
PoT program-of-thought	Python <code>decide()</code>	model-written code	framing
FG formula-given	probability ^a , action	model	—
G guardrail	quantities	exact solver	framing

^aIn F1, F2 and F4; F3 asks for the action only.

not knowing the rule from failing to apply it; newsvendor biases have been reported to persist when the formula is given [TODO: cite: Liu, Chen & Zhong 2025 (L32), preprint; sole-source exception pending].

Extract-then-solve guardrail (G). The model only extracts the item’s quantities (prior, likelihoods, costs, demand table and action labels) into the family’s JSON schema, and the family’s exact solver chooses the action. G is run on the canonical item and on its framed variant, and it has its own noise floor. An oracle control runs the solver on the true quantities.

Scale. P0 and G on the canonical items are run on all [TODO: param: items per family] items of each family, because the headline measure of information pricing needs equal and large test cells in F1 (Section 5.1). Every other condition and variant is run on the first [TODO: param: shared prefix per family] items of each family, and every paired comparison uses the items the two conditions share.

4.3. Parsing, failures and checks by construction

Answers are parsed against the condition’s schema. A parse failure is retried once; whether the retry changed the output is recorded, since under greedy decoding it usually does not. An answer that still cannot be parsed, a truncated answer, or, under PoT, a syntax error, exception, timeout or invalid action label, is scored as the item’s worst action in the primary analysis and excluded in a sensitivity analysis; both are reported, with the parse-failure rate per cell. A prompt template whose parse-failure rate in the pilot ([TODO: param: pilot items per family] items per family, excluded from

the main set) exceeds [TODO: param: pilot parse-failure limit] could be rewritten once, before the main run; no claim is read from the pilot.

Several properties hold by construction, and a violation is treated as a bug, not as evidence. They are checked before any outcome is read (Appendix Appendix B): the exact solver has zero regret and each heuristic behaves as its stratum requires (K1); the guardrail with oracle extraction has zero regret and no flips (K2); variants have identical expected-loss vectors (K3); mock models with fixed heuristics reproduce their analytically known regret (K4); weight-file hashes match the pinned values at the start of every job (K5); repeated runs in fresh processes agree, or their mismatch enters the noise floor (K6); the exact posteriors reproduce a unit slope in the probability regression (K7); the F4 and F1 balance cells are filled in every block (K8, K9); and oracle P2 and PoT answers have zero regret and zero decomposition terms, while the heuristic mock under FG reproduces its known regret (K10).

5. Pre-specified analysis

The metrics, tests and reading rules of this section were fixed before any model was run and are applied as written. The study is organized around two questions. The first is whether models pay for information exactly when it is worth its cost (Section 5.1). The second is where the loss of a direct decision arises: in reading the problem, in inferring a probability from it, or in turning that probability into an action (Section 5.2). The guardrail comparisons (Section 5.3) test the design answer that follows from them. The remaining measurements (Section 5.4) replicate, on multi-action families and pinned open weights, findings that have already been reported for other tasks and panels, and they are presented as replications.

Common rules. Items are the sampling unit, and every statistic is computed per (model, family) unless stated otherwise. Confidence intervals are two-sided at the [TODO: param: confidence level of every interval, B001]% level from the item bootstrap with [TODO: param: bootstrap resamples] resamples, whose seeds come from the registered block. Differences of two proportions use the Newcombe hybrid score interval [TODO: cite: Newcombe 1998, original-source exception pending] whenever either proportion rests on fewer than [TODO: param: Newcombe event threshold] events, and always for the value-of-information measure. Comparisons repeated

over families or transformations are corrected by Holm’s step-down procedure [TODO: cite: Holm 1979, original-source exception pending] at family-wise level $\alpha =$ [TODO: param: alpha]. Each claim is read per model as *supports*, *weakens*, *refutes* or *inconclusive* by the rules below, and the paper’s reading is the outcome for a majority of the panel actually run, that is, for more than half of it ([TODO: param: majority] of [TODO: param: panel size] models). Every model and family is reported, negative cells beside positive ones. A model whose calls could not all be completed within the job’s time limit after one resubmission is reported but not counted in the majority, and a claim is inconclusive if the parse-failure rate exceeds [TODO: param: parse-failure inconclusive rate] in more than half of its (model, family) cells.

5.1. Value-of-information discrimination

In F1, buying the test is the Bayes first action on half of the items of each stratum and a wrong action on the other half (Section 3.2). A decision-maker who prices information correctly buys the test on the first half and never on the second. The *value-of-information discrimination* of a model in a condition is

$$D = \Pr(\text{buys the test} \mid \text{testing is optimal}) - \Pr(\text{buys the test} \mid \text{testing is not optimal}), \quad (2)$$

estimated by the two sample proportions on the F1 items and bounded by the Newcombe interval. The exact solver has $D = 1$; a model that buys the test at a rate unrelated to its value has $D = 0$. D measures only whether the purchase tracks the value of the information; its two components, the over-testing rate (buying when testing is not optimal) and the under-testing rate (not buying when it is), are reported separately, and the direction of any mispricing is described rather than predicted.

Reading rule (C5). C5 holds that direct decisions misprice information. It is *supported* if, for a majority of models, D under P0 has an upper [TODO: param: confidence level of] bound below [TODO: param: C5 bound for P0] while D under the guardrail has a lower bound at or above [TODO: param: C5 bound for G]; *refuted* if, for a majority, D_{P0} has a lower bound at or above [TODO: param: C5 bound for P0]; *weakened* if D_{P0} is below [TODO: param: C5 bound for P0] for a majority but D_G misses [TODO: param: C5 bound for G] for a majority, so that the guardrail does not repair the pricing; and *inconclusive* if the interval of D_{P0} contains [TODO: param: C5 bound for P0] for a majority. C5 is

read only on the full F1 file ([TODO: param: items per family] items, half of them in each test cell). With fewer items the lower bound of D_G cannot reach [TODO: param: C5 bound for G] even for an error-free guardrail, so the rule could not be met; at a smaller scale C5 is recorded as inconclusive and D is reported without a reading. Reported with no rule: D under P1, P2, FG and PoT, and, in F1b, the number of tests bought minus the optimal number.

5.2. Regret decomposition

A P2 answer contains the model’s reading of the quantities $\hat{q}$, its stated probability $\hat{p}$ and its action $\hat{a}$. Let a_{read} be the exact solver’s action on the model’s own reading $\hat{q}$, and a_{inf} the Bayes action at the stated probability $\hat{p}$ under the costs in $\hat{q}$. The regret of $\hat{a}$ then splits exactly into three terms,

$$L(\hat{a}) - L^* = \underbrace{L(a_{\text{read}}) - L^*}_{\text{reading}} + \underbrace{L(a_{\text{inf}}) - L(a_{\text{read}})}_{\text{inference}} + \underbrace{L(\hat{a}) - L(a_{\text{inf}})}_{\text{action}}, \quad (3)$$

because the sum telescopes; each term is normalized by $L_{\text{max}} - L^*$ as in (1). The reading term is the loss a perfect reasoner would incur from the model’s reading alone; the inference term is the further loss of the model’s probability, given that reading; the action term is the loss of not acting on its own probability. In F3, which has no probability step, the inference and action terms are merged. The *reading share* of a (model, family) cell is its mean reading term divided by its mean total regret, estimated with the item bootstrap of the ratio of means. An answer whose $\hat{q}$, $\hat{p}$ or $\hat{a}$ cannot be parsed is excluded from the decomposition, and the exclusion rate is reported. On F1–F2 the action term is the loss-weighted counterpart of the belief–action incoherence rate of Section 5.4, and both are reported. The decomposition extends the two-term split of regret into a belief loss and an optimization residual [TODO: verify N1] by an explicit reading stage, which is where a guardrail can fail.

Reading rule (C6). C6 holds that most of the loss is not misreading. A cell whose mean total P2 regret has an upper [TODO: param: confidence level of every interval, B001]% bound below δ has no loss to locate and is marked *not applicable*. C6 is *supported* if, for a majority of models, the reading share has an upper [TODO: param: confidence level of every interval, B001]% bound below [TODO: param: C6 share threshold] in at least three of the applicable families (at least two if only three are applicable); *refuted* if, for a majority, its

lower bound is above [TODO: param: C6 share threshold] in at least two families; *weakened* if support holds in only one or two families; and *inconclusive* if more than half of the (model, family) cells are not applicable or have a reading-share interval that contains [TODO: param: C6 share threshold]. The signed inference and action terms are reported per cell.

Order sensitivity. A telescoping split depends on the order in which the stages are corrected: each later term absorbs its interaction with the earlier ones, so the primary reading share is the loss of misreading alone, with the model’s own inference and action held fixed. The primary order follows the pipeline, reading then inference then action. The action term is last in every order, because the model’s action is observed only after its own reading and inference. As a sensitivity with no reading rule, reading and inference are also split in the reverse order and the two orders are averaged, which is the two-player Shapley split [TODO: cite: Shapley 1953, original-source exception pending] of their sum. With a_{alt} the Bayes action at $\hat{p}$ under the item’s *true* costs, the reverse order assigns $L(a_{\text{alt}}) - L^*$ to inference and $L(a_{\text{inf}}) - L(a_{\text{alt}})$ to reading. Because $\hat{p}$ already embeds the model’s reading of the prior and the likelihoods, the reverse order isolates misread costs only, and the paper states this limit with the result. The Shapley reading share and its interval are reported for F1, F2 and F4; F3 has no probability and admits no second order. If the Shapley share falls on the other side of [TODO: param: C6 share threshold] for a majority of models, this is stated beside C6’s outcome, which it does not change.

5.3. Guardrail comparisons

Guardrail against the model deciding (C4). Three parts are tested for each model: (i) the guardrail’s mean regret is below P0’s in all four families and below P1’s in at least three, by the paired item bootstrap of the difference with Holm correction over families; (ii) the guardrail’s flip rate under framing minus its own noise floor has an upper [TODO: param: confidence level of every interval, B001]% bound at or below [TODO: param: C4 flip margin]; and (iii) its field-level extraction accuracy, with numbers correct within a relative tolerance of [TODO: param: extraction tolerance] and categorical fields exact, has a lower [TODO: param: confidence level of every interval, B001]% bound at or above [TODO: param: C4 extraction accuracy] in every family. C4 is *supported* if all three hold for a majority of models; *refuted* if, for a majority, the guardrail’s point estimate of regret is not below P0’s in at least three

families; *weakened* if (i) holds against P0 but not against P1 in at least two families, or if (ii) or (iii) fails for a majority in any family, in which case the paper reports where extraction fails; and *inconclusive* if the paired intervals contain zero in more than half of the (model, family) cells. The item-level exact-extraction rate and the cost per decision in tokens and server seconds are reported beside it.

Verified solver against model-written code (C4b). The guardrail and PoT are compared on the same items by the paired item bootstrap of $\text{regret}(G)$ minus $\text{regret}(\text{PoT})$, with Holm correction over the four families at level α , and by the difference of their flip rates under framing, bounded by the paired interval (Newcombe when either count is below [TODO: param: Newcombe event threshold]). C4b is *supported* if, for a majority of models, the guardrail’s regret is below PoT’s in at least three families and the flip-rate difference has an upper [TODO: param: confidence level of every interval, B001]% bound below zero; *refuted* if, for a majority, the guardrail’s point estimate of regret is at least PoT’s in at least three families; *weakened* if only one of the two parts holds for a majority; and *inconclusive* if the paired intervals contain zero in more than half of the (model, family) cells. If PoT matches the guardrail, the recommendation becomes to hand the computation to code, and the schema guardrail is one instance of it.

5.4. Replications and secondary metrics

The following measurements replicate effects already reported for direct LLM decisions and stated probabilities [3, 4, 5, 6]. Two of them repeat the closest prior designs on a new object. C1 (regret against an exact optimum) replicates the regret evaluation of Sun et al. [TODO: verify N1], which uses binary actions, on multi-action families with a costly-information option. C2(b) (belief–action incoherence) replicates the coherence finding of Yamin et al. [TODO: cite: Yamin et al. 2026 (L28), ICLR; bib entry pending a venues.csv row] with known costs, so that incoherence can be scored in expected loss.

Regret and traps (C1, C1’). C1 is *supported* if, under P0, the lower [TODO: param: confidence level] bound of mean normalized regret exceeds δ in at least three of the four families and the trap-minus-non-trap regret difference, pooled over families, has a lower bound above zero, both for a majority of models. It is *refuted* if, for a majority, the upper bound is below δ in at least three families, *weakened*

if the first part holds in only one or two families or the trap difference is not above zero, and *inconclusive* if the interval contains δ in more than half of the (model, family) cells. C1' asks whether wrong answers on trap items follow the named heuristic. In F1 and F3, where items have three or more actions, each wrong P0 answer on a trap item is compared with the share expected if wrong answers were spread uniformly, $1/(k - 1)$ for an item with k actions; the statistic is the observed heuristic share minus the mean expected share, pooled over F1 and F3. C1' is *supported* if its lower bound is above zero for a majority, *refuted* if its upper bound is at or below zero for a majority, *weakened* if it holds in only one of the two families, and *inconclusive* for a model with fewer than [TODO: param: C1-prime minimum wrong answers] wrong trap answers (and overall if that is true of a majority). Because a uniform spread is a weak baseline, two further expected shares are reported with no rule: that of the least-regret wrong action, and, on ordered action sets, that of the wrong action adjacent to the Bayes action.

Stated probabilities (C2). For C2(a), the stated F4 posteriors, clipped to [TODO: param: probability clip range] so that the logit is finite, are regressed as $\text{logit } \hat{p} = \beta_0 + \beta_1 \text{logit}(\text{prior}) + \beta_2 \log \text{LR} + \varepsilon$, where Bayes' rule gives $\beta_1 = \beta_2 = 1$. The fit uses only the items whose exact posterior lies in [TODO: param: C2(a) fitting window], a set defined by the item alone; on it the exact posteriors are never clipped and reproduce the unit slopes (check K7), whereas including extreme posteriors would bias the slope of an exactly Bayesian answer below one. The fit on all items and the paired difference between the model's slope and the exact posteriors' slope are reported as sensitivities. For C2(b), the *incoherence rate* in F1–F2 is the share of items on which the model's action differs from the Bayes action at its own stated probability under the item's true costs. C2 is *supported* if, for a majority of models, (a) β_1 's interval lies entirely below one and the mean absolute error $|\hat{p} - p^*|$ has a lower bound above [TODO: param: C2(a) MAE threshold], and (b) the incoherence rate has a lower bound at or above [TODO: param: C2(b) incoherence threshold]; *weakened* if only one of (a) and (b) holds; *refuted* if, for a majority, β_1 's interval contains one, the error's upper bound is below [TODO: param: C2(a) MAE threshold] and the incoherence rate's upper bound is below [TODO: param: C2(b) refutation bound]; and *inconclusive* if, for a majority, the incoherence interval contains [TODO: param: C2(b) incoherence threshold] or β_1 's interval is wider than [TODO: param: C2(a) CI-width limit]. The Brier score and the expected calibration error

([TODO: param: ECE bins] equal-mass bins, against outcomes drawn from the exact posterior with a registered seed) are reported without thresholds.

Invariance (C3). For each transformation T (framing, option order, paraphrase), the flip rate f_T under P0 is compared with the noise floor f_0 . C3 is *supported* if, for a majority of models, some transformation has an excess $f_T - f_0$ of at least [TODO: param: C3 excess] with a one-sided lower [TODO: param: confidence level of every interval, B001]% bound above zero, with Holm correction over the three transformations; *refuted* if every transformation has an upper bound of the excess below [TODO: param: C3 excess]; *weakened* if the excess holds only for option order, a known format effect [TODO: cite: Pezeshkpour & Hruschka 2024 (L11), Findings of NAACL; bib entry pending a venues.csv row]; and *inconclusive* if, for a majority, the transformation with the largest excess has a point excess of at least [TODO: param: C3 excess] but a lower bound not above zero, or if the intervals neither exclude [TODO: param: C3 excess] nor lie below it. Because greedy decoding leaves a floor near zero, the difference uses the Newcombe interval whenever either rate rests on fewer than [TODO: param: Newcombe event threshold] events. C3 also serves as the invariance test for the guardrail and for PoT above.

Baselines and secondary metrics. Every table reports, beside the models, the exact solver, each trap heuristic, a uniformly random action and every constant-action policy (for example, always testing in F1 or always declining in F4), together with each model’s share of each action, so that a model’s regret can be read against the best policy that ignores the item. Accuracy (the share of optimal actions), the parse-failure rate, and tokens and server seconds per decision are reported per cell.

6. Results

[Intent: Report each claim’s outcome as recorded, headline first: VOI pricing (C5), then the regret decomposition (C6), then the guardrail against direct, reasoning and program-of-thought (C4, C4b), then the replications (C1, C1’, C2, C3); per model and pooled.]

- 6.1. *Do models price information? (C5)*
- 6.2. *Where the loss arises: reading, inference or action (C6)*
- 6.3. *Extract-then-solve against direct, reasoning and program-of-thought (C4, C4b)*
- 6.4. *Replications: regret, traps, probabilities and invariance (C1, C1', C2, C3)*

7. Discussion

[Intent: Implications for designing decision support around LLMs (what to hand to the model, what to hand to a solver); limitations (open-weight panel only, synthetic items, English prompts, decomposition order); threats to validity; how the framing changes under each recorded outcome, as stated in ARGUMENT.md.]

7.1. *Implications for decision support design*

7.2. *Limitations and threats to validity*

8. Conclusion

[Intent: Restate the question, the recorded outcomes of C5 and C6, the guardrail result, and what a practitioner should do.]

Data and code availability

[Intent: Generators, solvers, prompt templates, harness, item files (released after all runs), raw model responses, pinned model identifiers and result files, deposited on acceptance with a persistent identifier.]

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies in the writing process

[Intent: Owner’s statement on AI-assisted writing and coding, in Elsevier’s template wording.]

Appendix A. Prompt templates, schemas and reference solvers

[Intent: Give the fixed prompt templates per family and condition, the guardrail and P2 answer schemas, the formula-given sentences and the reference `decide()` functions.]

Appendix B. Checks by construction and generator diagnostics

[Intent: Report checks K1–K10 and the generator diagnostics (separation, strata and F1 test-action balance).]

References

- [1] H. Raiffa, R. Schlaifer, Applied Statistical Decision Theory, Studies in Managerial Economics, Division of Research, Graduate School of Business Administration, Harvard University, Boston, 1961.
- [2] S. G. Pauker, J. P. Kassirer, The threshold approach to clinical decision making, New England Journal of Medicine 302 (20) (1980) 1109–1117. doi:10.1056/NEJM198005153022003.
- [3] O. Macmillan-Scott, M. Musolesi, (ir)rationality and cognitive biases in large language models, Royal Society Open Science 11 (6) (2024) 240255. doi:10.1098/rsos.240255.
- [4] G. Suri, L. R. Slater, A. Ziaee, M. Nguyen, Do large language models show decision heuristics similar to humans? A case study using GPT-3.5, Journal of Experimental Psychology: General 153 (4) (2024) 1066–1075. doi:10.1037/xge0001547.

- [5] Y. Chen, S. N. Kirshner, A. Ovchinnikov, M. Andiappan, T. Jenkin, A manager and an AI walk into a bar: Does ChatGPT make biased decisions like we do?, *Manufacturing & Service Operations Management* 27 (2) (2025) 354–368. doi:10.1287/msom.2023.0279.
- [6] L. Qiu, F. Sha, K. Allen, Y. Kim, T. Linzen, S. van Steenkiste, Bayesian teaching enables probabilistic reasoning in large language models, *Nature Communications* 17 (2026) 1238. doi:10.1038/s41467-025-67998-6.